

CHAPTER 18

Telling a Story with Graphs: How to Apply Narrative Visualization Strategies and Visualization Techniques in Journalistic Storytelling

Zhao Peng

Introduction

Storytelling enables us to arrange our experiences in our memories, convey meaning, and construct and structure our visions of the world we strive to know (Vázquez-Herrero & de Haan, 2022). Journalists rely on storytelling to represent reality and communicate it to the public. The irruption of information technologies has placed journalism in rapid changes and renovated the storytelling formats and models. With the empowerment of digital technologies, journalists have started to be more creative and strategic in presenting their stories, incorporating moving images, sound, videos, interactive data visualization, animations, infographics, multimedia elements, and immersive technology in their storytelling.

Of these digital storytelling forms, data journalism, or data storytelling, has gained prominence in both big and small news media organizations. Data journalists heavily rely on large sets of data to write news stories and use data visualizations to communicate the final story (Stalph & Heravi, 2023; Weber & Rall, 2012). Data journalists incorporate static or dynamic graphics to tell complex stories, provide insights, reveal trends and patterns, make stories more

Z. Peng (✉)
Department of Journalism at Emerson College, Boston, MA, USA
e-mail: zhao_peng@emerson.edu

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accessible, and enable audiences to explore datasets (Rodríguez et al., 2015). Nearly 99% of projects nominated by Data Journalism Awards between 2013 and 2016 included data visualizations, which indicates that data visualizations have become an integral part of data-driven stories (Loosen et al., 2020). Storytelling with data visualizations is not simply embedding bar charts within texts. Without storytelling techniques, data visualizations would not be able to achieve the goal of successful storytelling. Storytelling with data visualization differs from storytelling with text in essential ways. For example, stories in text are typically presented either in a series of linear sequences or in an inverted pyramid structure, which is strictly organized by the author. In contrast, stories in data visualization can be organized in a linear sequence (author-driven) but can also be presented in a non-linear sequence and interactive, inviting audiences to discover the path of the story (reader-driven) (Rodríguez et al., 2015; Segel & Heer, 2010).

Regarding data visualizations in news stories, journalism scholars focused on data journalism practices (Engebretsen et al., 2018; Heravi & Lorenz, 2020), interactive features (Appelgren, 2018), data visual elements (Stalph & Heravi, 2023), and solutions for historically archiving interactive data visualizations (Heravi et al., 2022). In comparison to studies from the fields of computer science, infographics, and human-computer interaction, studies from journalism scholarship rarely talk about or study the strategies and techniques of incorporating data visualizations in journalistic storytelling. Building on the literature on narrative visualization and visualization rhetoric, this chapter aims to summarize and explain how narrative visualization design strategies and visualization techniques are being used in data visualization to improve journalistic storytelling and discuss how data and digital journalists can benefit by using such techniques to enhance audiences' interpretations of data visualization stories.

From Inverted Pyramid to Innovative Storytelling

Advances in technology have pushed the evolution of journalism, bringing along negotiations of journalistic rules, norms, values, practices, and storytelling formats (Song, 2018). Storytelling in journalism refers to the narrative structure of news (Emde et al., 2016). The affordances of new technologies enabled journalists to pursue innovative modes of storytelling, which shifted from non-participatory and distancing to a more personal and interactive narrative mode (Wahl-Jorgensen & Schmidt, 2019). Although new modes of storytelling challenge traditional journalism norms and practices, it also presents new opportunities for connecting with readers.

The inverted pyramid (IP) structure has been a trademark and standardized routine in journalism, for its emphasis on objectivity and constraints on journalists' own values and emotions (Shoemaker & Reese, 1996; Nee & Santana, 2022). The rise of IP is due to the wide adoption of the telegraph at the beginning of the nineteenth century (Canavilhas, 2007). Back in that time,

journalists sent news in paragraphs via telegraph, in which information was sent in descending order of importance (Pöttker & Starck, 2003). Such information delivery created the specific journalistic storytelling structure-inverted pyramid, which is contrary to the chronological narrative structure (Norambuena et al., 2020). Structuring information in IP style is important for stories characterized by high news value as the IP structure highlights fact-based reporting, objectivity, and neutrality rather than journalistic interpretation in news writing (Nee & Santana, 2022). Given this advantage, the IP structure is not only popular in print journalism but also adopted by broadcast journalism and, later, online journalism.

Despite the IP's dominance in news writing, the emergence of new technology has prompted scholars and journalists to question and re-examine the efficacy of IP in news delivery (Ytreberg, 2001). Scholars criticized the IP structure for being too strict and formulaic and only representing one end of a story and leaving the other end open (Kosara et al., 2023). Reading IP-structured stories also relies on significant cognitive resources as audiences need to access previously stored knowledge to be able to finish the story (DeAngelo & Yegiyan, 2019). While the IP structured stories require considerable cognitive capacity, available resources are insufficient for audiences to fully understand the entire story (Kleemans et al., 2018). Machill et al. (2007) conducted an experiment to test the impact of two narrative structures—IP and linear narrative, on audiences' news understanding and engagement. Results showed that linear narrative-structured stories were perceived as more effective and engaging in conveying information.

The introduction of digital media and the Internet brought along new modes of storytelling. News storytelling still features a non-linear narrative form. However, different from the IP structure, storytelling in the age of digital media allows audiences to experience and explore the story in a variety of narratives (Pavlik & Pavlik, 2017). News stories are no longer limited to text or images. Journalists can take advantage of digital media's unique characteristics and integrate a plethora of media formats, including audio, video, data visualizations, graphic animations, virtual/augmented reality, computer algorithms, and games, into news storytelling, helping convey information in an interactive, immersive, and engaging way (Pavlik & Pavlik, 2017; van Krieken, 2018). Some news media developed game formats for news stories that allow audiences play against a computer algorithm to explore and understand the people, events, and contexts in the news event (Pavlik & Pavlik, 2017). Virtual reality and augmented reality are also being experimented with in online news stories to create a more immersive experience.

Of new emerging journalistic storytelling, data storytelling, which involves using data to uncover insights and tell compelling stories, is becoming increasingly prominent in digital journalism with wide adoption by small and large media outlets. Data storytelling is not simply conveying information in numbers or charts, but telling a story that is comprehensive, memorable, and engaging. Applying data storytelling principles and techniques can improve

audiences' recall and engagement of information presented in visualizations (Liem et al., 2020). Therefore, for the rest of this chapter, I will focus on the strategies and techniques that can help journalists improve the effect of data storytelling.

Data Storytelling and Data Visualization

Data journalism is a type of journalism where reporters apply the skills and knowledge of data science to journalism (Howard, 2014). The data science skills and knowledge include gathering and validating data as a source, using statistical knowledge to interrogate data, and visualizing data to support the news story. Many journalism practitioners and scholars view data journalism as a way to improve journalism to be more systematic, accurate, and trustworthy (Borges-Rey, 2017; Boyles & Meyer, 2017). Data journalism also blurred the boundary of the professional domain, as journalists, data scientists, designers, and computer scientists are all playing significant parts in data story creation. Data stories, therefore, are multimodal artifacts that integrate together texts, visuals, codings, and design into a coherent whole (Weber et al., 2018).

Data storytelling is a structured approach by which authors arrange factual pieces in a meaningful order and represent data in a visual format to communicate insights and facilitate understanding (Edmond & Bednarz, 2021; Ojo & Heravi, 2017). The process of creating a data story comprised three major phases: exploring data, making a story, and telling a story. Exploring data refers to activities of exploring and analyzing the raw data that constitutes the content of a story (Lee et al., 2015). The purpose of exploring data is to collect data excerpts that contain insights, variations, or externalization of data. In the story-making process, journalists need to sequence the collected data excerpts into an interesting and compelling storyline. Not only are journalists just ordering data excerpts, they also need to build logical connections, develop flow, and formulate the main message (Lee et al., 2015). Through this phase, journalists should be able to communicate clearly with the story about how the data excerpts are logically related and what all these mean in the big context. Lee et al. (2015) proposed that the final phase includes transforming the abstract content into visual representations, sharing the story, and receiving feedback from audiences.

Data visualization is defined as any methodology that can represent data in a visual format that showcases insights and facilitates understanding (Engebretsen et al., 2018; Kirk, 2012). Different from information graphics, data visualization is deeply rooted in quantified facts (Friendly, 2008) and uses geometrical shapes and colors to map out data values and relations (Engebretsen et al., 2018). A well-designed data visualization not only catches audiences' attention but also cuts through the clutter of a complicated story to get right to the point. As advanced technology enabled data visualizations to have an interactive and dynamic interface, the role of data visualization gradually

extended to storytelling. Lopezosa et al. (2023) posited that data visualization occupies "a prominent place in storytelling" (p. 315).

Data visualizations serve as a potent tool and provide multiple functions for news storytelling (Engebretsen et al., 2018; Lopezosa et al., 2023). If using data visualizations as a central element in a story, it can emphasize a specific journalistic angle. When claims in a news story need facts to back them up, data visualizations can provide empirical evidence. Data visualizations also enable audiences to explore the dataset and filter out information that is personally relevant. Engebretsen et al. (2018) also pointed out that employing exploratory data visualizations in storytelling can open the possibility for audiences to create many different stories or discover unseen stories.

However, including data visualizations to improve storytelling is not simply throwing charts in a news story. Data visualizations should assist in the unfolding of a story, which keeps audiences engaged but does not disrupt the pace and structure. Telling a compelling news story requires authors to understand the narrative structure that is familiar to the audience (Rayfield, 1972). It was found that audiences will easily recognize and accept a story if it possesses a certain structure, which is characterized by some complexity. Completely explorative visualizations without guidance are not considered a data story as there are no annotations or explanations to help audiences capture the intended message (Lee et al., 2015). In addition to structure, the techniques used in data stories also play an important role in conveying the intended story and affecting the interpretation of the audience. The rest of this chapter focuses on two major topics—the narrative visualization structure and visualization techniques—to discuss and conclude how previous scholars have discovered the optimal way to present data stories.

Narrative Visualization Structures and Other Structures

Narrative visualization is defined as "the process by which the author achieves the goals of data visualization using narrative techniques" (Edmond & Bednarz, 2021). In journalism, narrative visualization is a more novel concept to academics than to practitioners: news media outlets have been integrating dynamic visualizations in news stories for a long time, whereas academic discussion on this topic remains relatively rare (Howard, 2014). In comparison, scholars from engineering, informatics, computer science, and HCI have been studying narrative visualization from the perspectives of narrative structures, tactics, approaches, techniques, and tools. Of all narrative visualization elements, the narrative structure serves as a framework on which authors build, order, and present the narrative to audiences. A story's narrative structure can affect audiences' perception, understanding, and memory of it (Yang et al., 2022). Visual design considerations also rely on the narrative structure of a story. Therefore, drawing on the literature from the abovementioned disciplines, I will discuss

how journalists can enhance the efficacy of data storytelling by incorporating narrative visualization structures or other structures into data-informed stories.

Narrative Visualization Structures

The structure of narrative visualization lies in the spectrum of author-driven and read-driven approaches (Segel & Heer, 2010). The author-driven approach, which is optimum for storytelling and efficient communication, follows a strictly linear structure through visualization and messaging and allows no interactivity. In contrast, the reader-driven approach has a high degree of interactivity and no strictly structured narrative path. In reader-driven stories, data visualizations function as a tool that gives audiences maximum freedom to explore and analyze the data and story (Weber et al., 2018). Most of the time, news stories use a combination of author-driven and reader-driven approaches, falling between the two sides of the spectrum. Journalists need to strike a balance between leaving room for interactivity and laying out a structured narrative path.

Hybrid combinations of two approaches are most seen in online news stories. In the hybrid structure, the main story is provided by the author, but audiences also have the freedom to explore the data in a certain part of the story (Weber et al., 2018). Segel and Heer (2010) identified three typical hybrid combinations, each of which has different permutations and emphasis and works for different types of stories.

The Martini Glass structure begins with the author-driven approach, using questions, observations, or written articles to construct the author's predetermined sequences (Segel & Heer, 2010). Once the author-driven part is finished, the visualization opens up to the reader-driven part, inviting audiences to explore the data and interact with the display. Authors can use questions, observations, paragraphs, or articles in the author-driven part and use highlighting, filtering, or path-choosing in the reader-driven part. The Martini Glass structure is often encountered in interactive visualization news stories (see Fig. 18.1).

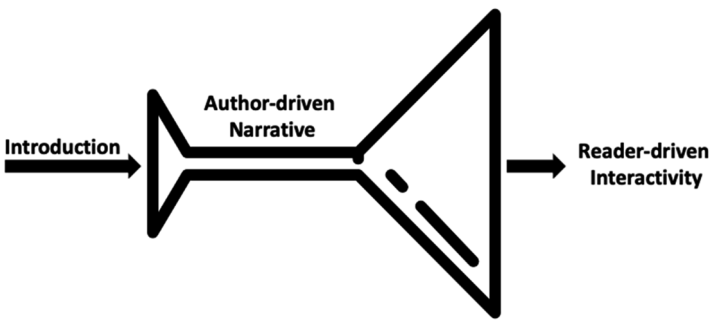


Fig. 18.1 Martini glass structure

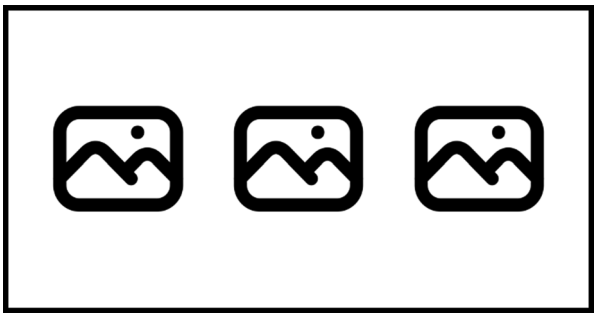


Fig. 18.2 Interactive slideshow structure

The Interactive Slideshow structure follows a typical slideshow format but allows audiences to interact with some slides in the middle before moving forward to the next slide (Segel & Heer, 2010). Interestingly, in the interactive slideshow, each interactive slide is structured in Martini Glass style: beginning with an author-intended narrative and then following a visualization with which audiences interact. The interactive slideshow is recommended to be used for multi-dimensional datasets or complex stories. Each slide can navigate audiences through the dimensions of a dataset and help them obtain a deeper understanding of each dimension through interaction. Audiences are given enough time to digest one slide's information before clicking on the next slide. For complex stories, the slideshow format provides audiences with a clear roadmap that enables audiences to distinguish different segments within a big story (see Fig. 18.2).

The Drill-Down Story structure is more reader-driven oriented (Segel & Heer, 2010). The Drill-Down Story visualization comprises a major theme and several particular sub-sections that let audiences click to reveal more additional details and backstories (see Fig. 18.3). Although the Drill-Down Story is more reader-driven inclined compared with the above two structures, it still needs authors to decide which sub-section on the visualization includes more additional details or backstories.

Other Structures

The Stack of Blocks

The Stack of Blocks is another storytelling structure that has been used by data journalists (see Fig. 18.4). The Stack of Blocks structure contains a beginning, middle, and end. The beginning provides the introduction to the story and predicts the middle, the middle is divided into sections by sub-topics in a logical order, and the end reiterates the main point and strengthens audiences' memories (Fry, 2009; Heravi, 2022). Different from the traditional narrative structure, the Stack of Blocks does not have a dramatic arc. In contrast to drawing attention to the character's journey and the chronology of the events, the Stack of Blocks put emphasis on the discussion of each sub-topic within the

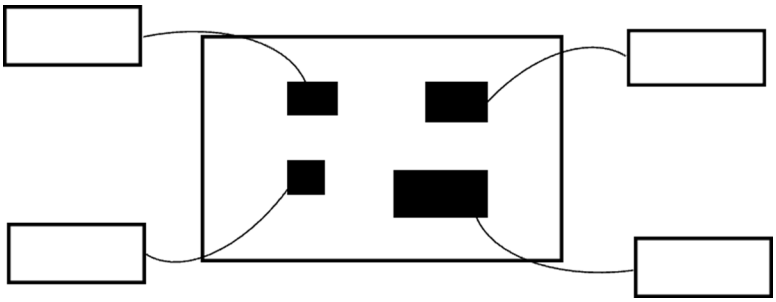
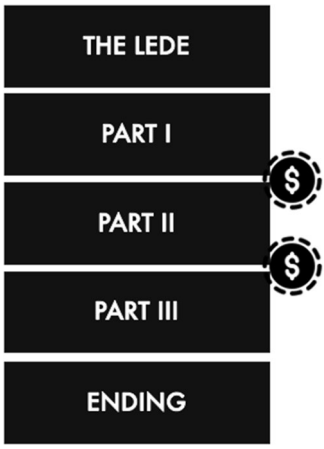


Fig. 18.3 Drill-down story structure

Fig. 18.4 The stack of blocks structure



whole story in a logical order, which is advisable for data journalism. In Fig. 18.4, the gold coin between two parts indicates the place to insert data visualizations.

The Water Tower Structure

After analyzing 118 data-driven stories, Heravi (2022) proposed a new structure—Water Tower structure, which fits many data stories. In the introduction part, the water tower structure takes the Inverted Pyramid structure and includes the key facts. When moving to the middle part, it takes the Stack of Blocks structure's stem, grouping information by subjects and arranging them in a logical manner. The end also follows the Stack of Blocks structure, cementing the end into audiences' memories. Overall, the Water Tower structure is an improved version of the Stack of Blocks structure.

The Lede in the Water Tower structure follows the Inverted Pyramid structure and presents the most important facts as the hook to attract the audience's attention. As audiences move down to the middle of the story, the Water Tower structure takes them deeper into the 5Ws and 1H by discussing relevant

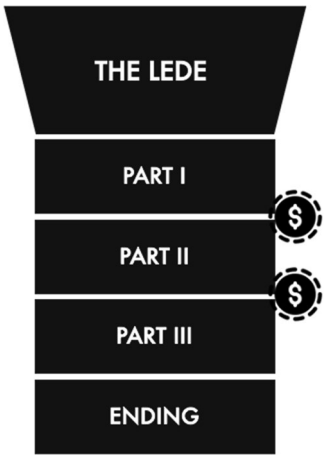


Fig. 18.5 The water tower structure

sub-topics. The coin sign in Fig. 18.5 indicates the place where authors can insert data visualizations to keep audiences stay interested in the story.

Visualization Techniques

Successful storytelling and engaging narrative visualization cannot be achieved without incorporating visualization techniques. Visualization techniques can influence audiences' experiences of interacting with an online article and understanding a story. Scrollytelling is a storytelling technique combining visuals and texts that change as audiences scroll through an online article. It is recommended for in-depth stories and can use a linear or elastic narrative structure. Annotations are important for guiding attention and conveying core messages. Visualization rhetoric is used to prioritize certain interpretations, and data visualization designers should create unambiguous designs. This section focuses on discussing these three major visualization techniques which have been recommended to use in data-driven stories.

Scrollytelling

Scrollytelling, which combines the terms "storytelling" and "scrolling", refers to a storytelling technique to make visuals and texts change as audiences scroll through an online article (Seyser & Zeiller, 2018; Oesch et al., 2022). A typical scrollytelling article starts with full-screen photography and progresses to the next element as audiences scroll vertically or seldom horizontally (Seyser & Zeiller, 2018). Although a large number of visual elements are integrated into scrollytelling articles, text is still the backbone and provides major content. The scrollytelling technique is recommended for in-depth stories that cover complicated topics and plenty of details, such as investigative journalism or data-driven journalism.

The scrollytelling technique suits either linear or elastic narrative structure; in other words, it can be used for either more author-driven or hybrid narrative visualization structure (Seyser & Zeiller, 2018). For an elastic narrative structure, the author still needs to have a predetermined order laid out for audiences but can branch off at specific stages of the story. These branches allow audiences to delve deeper into the story. Authors can also follow the structure of interactive slideshows or drill-down story: deconstruct a complex story into chapters, each of which follows an elastic structure. One caveat of using scrollytelling technique in a hybrid structure is that the author builds too many branches for audiences to find their way back to the main thread. Therefore, the author should ensure audiences can always go back to the core story no matter which branch they go to for more details.

There are five scrollytelling techniques in total: animated transitions, graphic sequence, pan-and-zoom, show-and-play, and moviescroller.

Graphic Sequence

In a graphic sequence, images stay fixed in the background. As audiences scroll, text blocks move over the image, and the current image will transition into the next one with fading animation (Oesch et al., 2022) (see Fig. 18.6). This technique is recommended for highlighting certain parts of a visualization step by step. To keep the design consistent and clean, the image should be kept the same throughout the scroller.

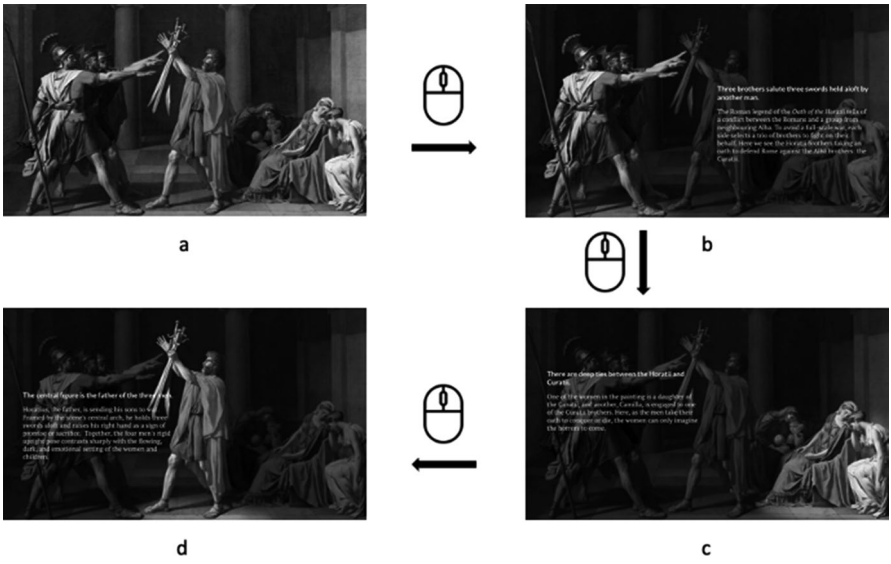


Fig. 18.6 Graphic sequence scrollytelling technique. As users scroll, text blocks move over the image, and different parts of an image are highlighted

Animated Transitions

In animated transitions, images or charts stay fixed as text blocks scroll alongside (Oesch et al., 2022). As one text block scrolls up, the image or chart transitions into another with morphing animation. This technique works well for building up complex data visualizations in a stepwise fashion, adding each component stage by stage (see Fig. 18.7). It can also be used to show different aspects of a dataset by dismantling and regrouping symbols of a unit in a data visualization.

Pan-and-Zoom

In pan-and-zoom transitions, one full-screen image with certain sections marked or highlighted for audiences (Oesch et al., 2022). As the audience scrolls, the image will zoom in to and enlarge one of the marked sections, displaying the necessary details of that section (see Fig. 18.8). Scrolling triggers the image to zoom or pan. The speed of the zooming or panning movement depends on how fast audiences scroll. The pan-and-zoom technique is often used to help audiences navigate through maps or zoom in on different parts of an image.

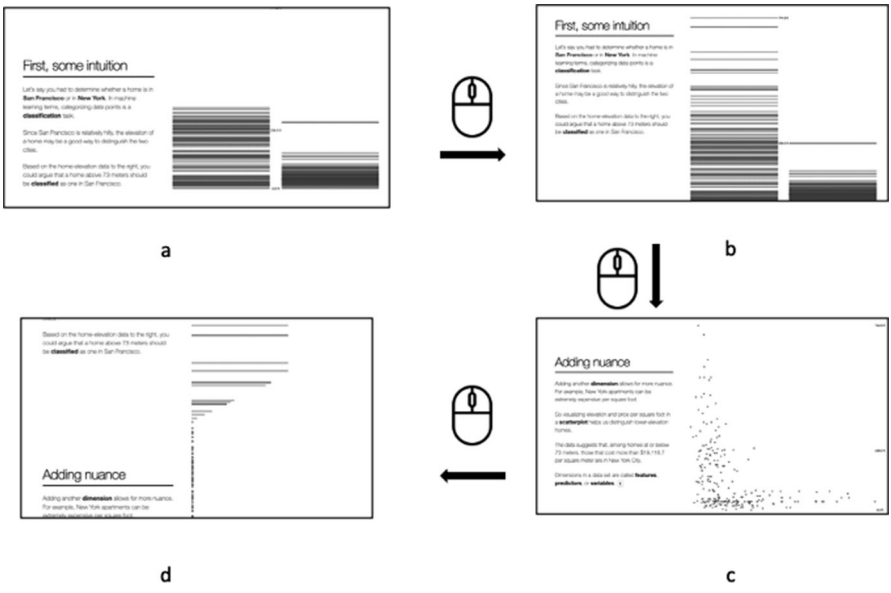


Fig. 18.7 Animated transitions scrollytelling technique. Text and image blocks stay in a fixed area. As text blocks scroll up, the image or visualization transitions into another with morphing animation

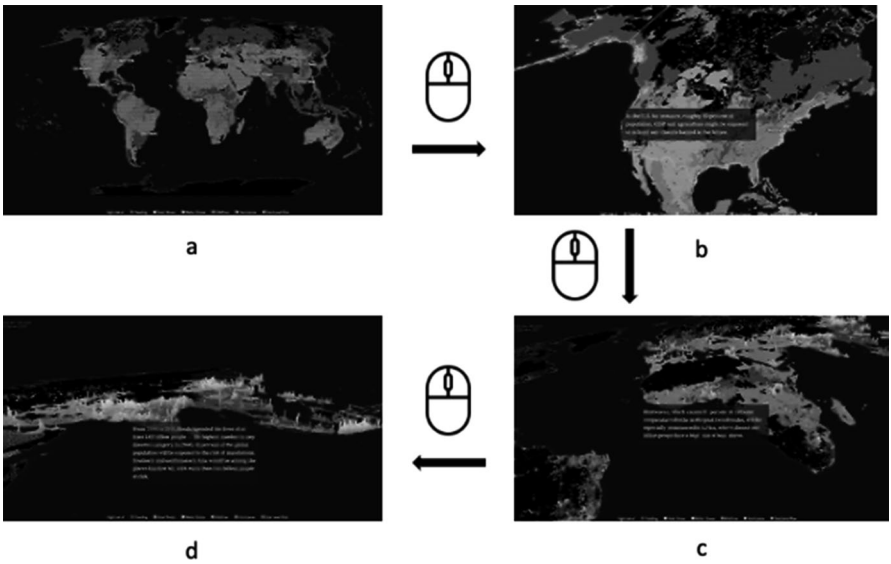


Fig. 18.8 Pan-and-Zoom scrollytelling technique. As users scroll, the map will zoom in to and enlarge one of the marked sections, displaying the necessary details of that section

Moviescroller
In moviescrollers, audiences scroll through a continuous sequence of images to create the moving effect (see Fig. 18.9). Moviescroller is often used to track shots in 3D models.

Show-and-Play
Compared with the other four scrollytelling techniques, show-and-play is the least interactive technique. In the show-and-play, both textual and visual elements are placed in a predetermined order. The visual elements could be GIFs or videos, which will automatically play in a loop when audiences scroll to the part. Play-in-a-loop will stop when audiences scroll that visual element away.

Annotations Techniques

Annotations are often overlooked in data-driven storytelling, but they play a crucial role in conveying authors' core messages. By drawing attention to particular areas of a graph, annotations can help viewers better understand the data. Prior data visualization scholars also emphasized the importance of textual and graphical annotations in helping audiences capture the core messages in data-driven stories (Choe et al., 2015; Lee et al., 2015). Annotations are not only referring to textual commentaries that are added to a chart (Ren et al., 2017). An annotation can be any kind of superstructural elements that assists

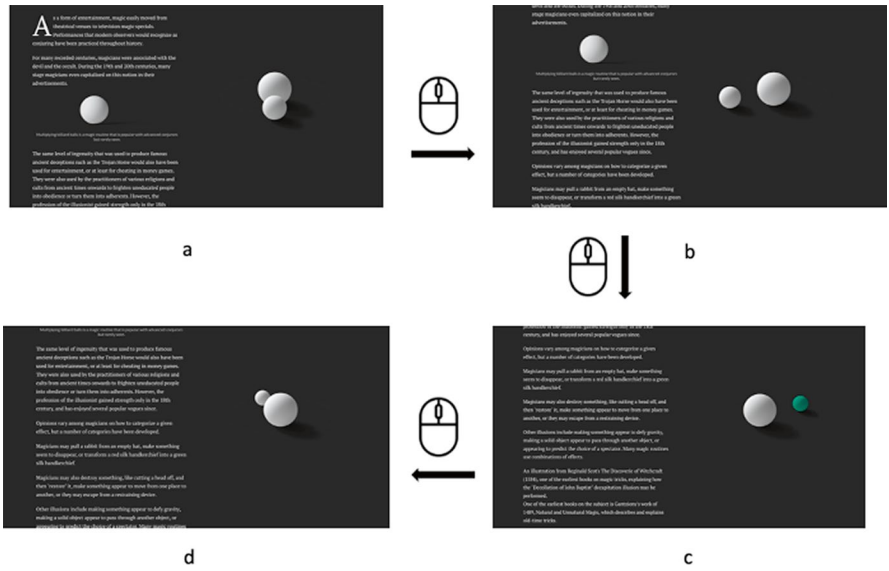


Fig. 18.9 Moviescroller technique. Users scroll to trigger the moving and changing of the small white ball in the image

authors present insights and facilitate audiences' understanding and interpretation (Kirk, 2012; Marshall, 1997). Ren et al. (2017) summarized four forms of visual annotations that had been used in data-driven stories: text, shapes, highlights, and images. Different forms of annotations serve different purposes in data visualization.

Text

Data-driven text annotations can be used to indicate the values of chart elements (Ren et al., 2017). For example, in Fig. 18.10, the percentages of both darker and white points in each circle point are presented in data-driven text annotations in the dot pie chart. When only a limited number of elements are annotated in a visualization, the purpose is to draw audiences' attention to them before seeing other elements that do not have annotations. Non-data-driven text annotations are used to provide information context, orientation, or editorial comment (see Fig. 18.10). The typography in annotations could also help convey meaning or facilitate understanding, which will be further discussed in the linguistic rhetoric section.

Shapes

Shapes, such as rectangles, brackets, ellipses, arrows, lines, and speech bubbles, can also be used as annotations (Ren et al., 2017). Authors can use rectangles or ellipses to enclose chart elements, thereby prompting the viewer to recognize the significance of these elements or to contrast them with other outside

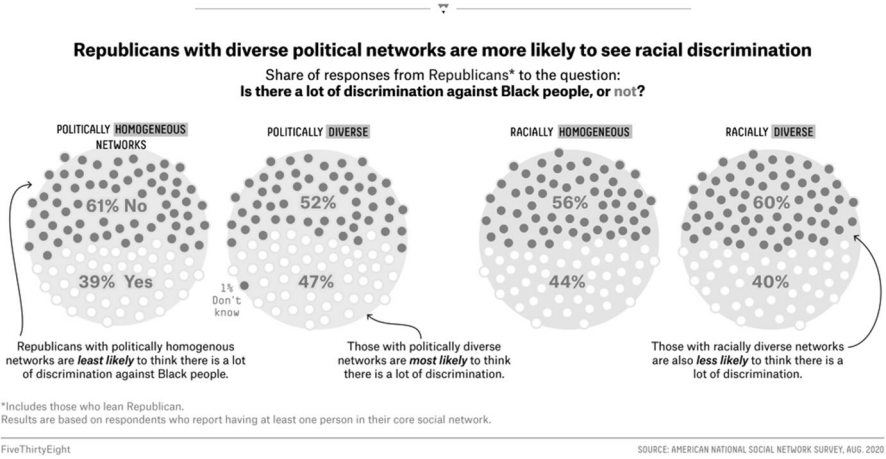


Fig. 18.10 An example of using data-driven text to indicate data values and using non-data-driven text to provide information context context. Source: Koerth & Thomson-DeVeaux (2021)

elements. Arrows and symbols can be used to guide the audience's attention to a particular element. Shapes annotations could also be data-driven. For example, in a scatter plot, authors can add a trendline to indicate the underlying trend of two variables.

Highlights
Authors can also use highlights to emphasize the importance of the target. The highlight can be achieved by changing the element's size, stroke, or color.

Images
Authors can add images and icons to the target to provide extra information and emphasize its importance. Images and icons can be distinguished by their size, opacity, or saturation, as well as their position in relation to the target. For example, in Fig. 18.11, authors used the images of senators and house representatives to indicate who reported a trade of stock or another financial asset by themselves.

Visualization Rhetoric Techniques

Given the interactive and explorative nature of narrative visualizations, which might lead to multiple interpretations on the audiences' side, visualization scholars have gone beyond studying the clarity and efficiency of information visualizations and started to look into how design strategies and rhetorical techniques can make certain interpretations more probable (Hullman & Diakopoulos, 2011). Visualization rhetoric refers to "the set of processes by which intended meanings are represented in the visualization via a designer's choices and then shaped by individual end-user characteristics, contextual

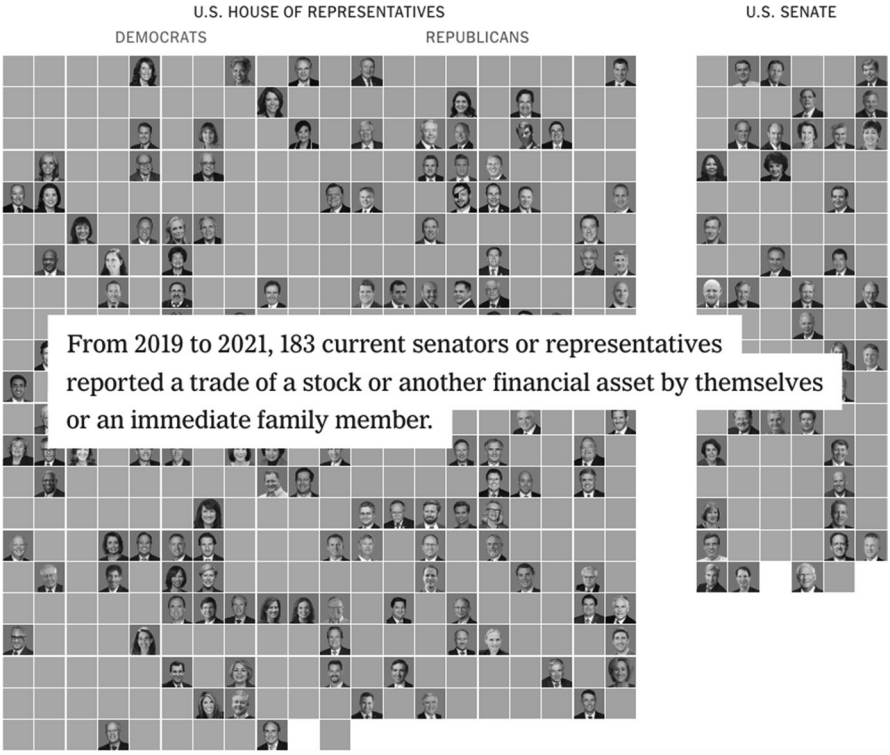


Fig. 18.11 An example of using images to provide extra information. Source: Kelly, Playford, Parlapiano, & Uz (2022)

factors involving societal or cultural codes, and the end-user's interaction" (Hullman & Diakopoulos, 2011, p. 2232). Prior scholars found that design elements that did not directly communicate the designers' intended meaning could bias audiences' interpretations of the designers' original meaning (Norman, 1999). This is because implicit meaning encoded in visual design may interact with audiences' knowledge and understanding of conventions, which subsequently influence their interpretation. Due to the bias caused by visualization design, data visualization designers should create unequivocal designs in which the intended meaning is transparent and graphs are absolutely unambiguous (Kosslyn, 2006). To achieve this goal, data visualization designers should be equipped with knowledge of how to use certain design elements strategically to prioritize certain interpretations and skills of designing narrative visualizations to communicate layered meanings without ambiguity.

Mapping Rhetoric

Mapping rhetoric refers to the manipulation of information presentation through data-visual transmission functions and the constraints that decide how

information is converted to visual elements (Hullman & Diakopoulos, 2011). Obscuring can occur by introducing noise into representations at the perception level, such as unnecessary size transformations. Visual elements that are too small for judgment, oversizing, or obscure the actual position of the value on an axis will all disrupt the audiences' interpretation of the visualization. Besides obscuring, Hullman and Diakopoulos (2011) also introduced three mapping rhetoric techniques that can be used in facilitating audiences' decoding and understanding of visualizations.

Visual Metaphor

In linguistics, metaphors are used to help audiences understand a primary and abstract concept by associating it with a secondary concept that is familiar and tangible to audiences (Borgo et al., 2012). Visual metaphor is a technique where a primary and abstract concept is conveyed by making good use of visual elements that are familiar to audiences. For example, computers use an icon of a paper folder, which is similar to the file folder in the real world, to teach users how to use it to store and organize files. Besides facilitating understanding, visual metaphors can also draw audiences' attention and strengthen the memory of presented information.

In data visualization stories, authors can also use the visual metaphor technique to convey implicit and abstract meanings and strengthen the memory of visualized content. More concretely, authors can use spatial mappings, typographic mappings, and color mappings (Hullman, 2020). For example, in election-themed stories, authors can use blue and red font colors to represent the republican and democratic parties.

Classification

Classification can be achieved through the categorization of data based on size, position, or color. Furthermore, the utilization of typographic manipulation of font sizes, styles, and equations of significance, presented in a legend-like format, can be used to emphasize certain values and thus classify the information within a visualization. This type of classification can be used to demonstrate clusters of priority or importance by making certain points stand out. Additionally, it can also help to organize data into easily understandable sections. For example, in Fig. 18.12, different shades and enclosed circles are employed to categorize 50 states that were required to get a gubernatorial recall. In this way, classification can be used to highlight patterns and trends contained within a dataset that may not be visible at first glance.

Redundancy

It has been observed that employing redundancy techniques can be beneficial in emphasizing homogenous values or visual marks (Hullman & Diakopoulos, 2011). This method can involve the utilization of identical labels or the disaggregation of values with minimal variance. Additionally, employing the redundancy technique can highlight certain information in a visualization.

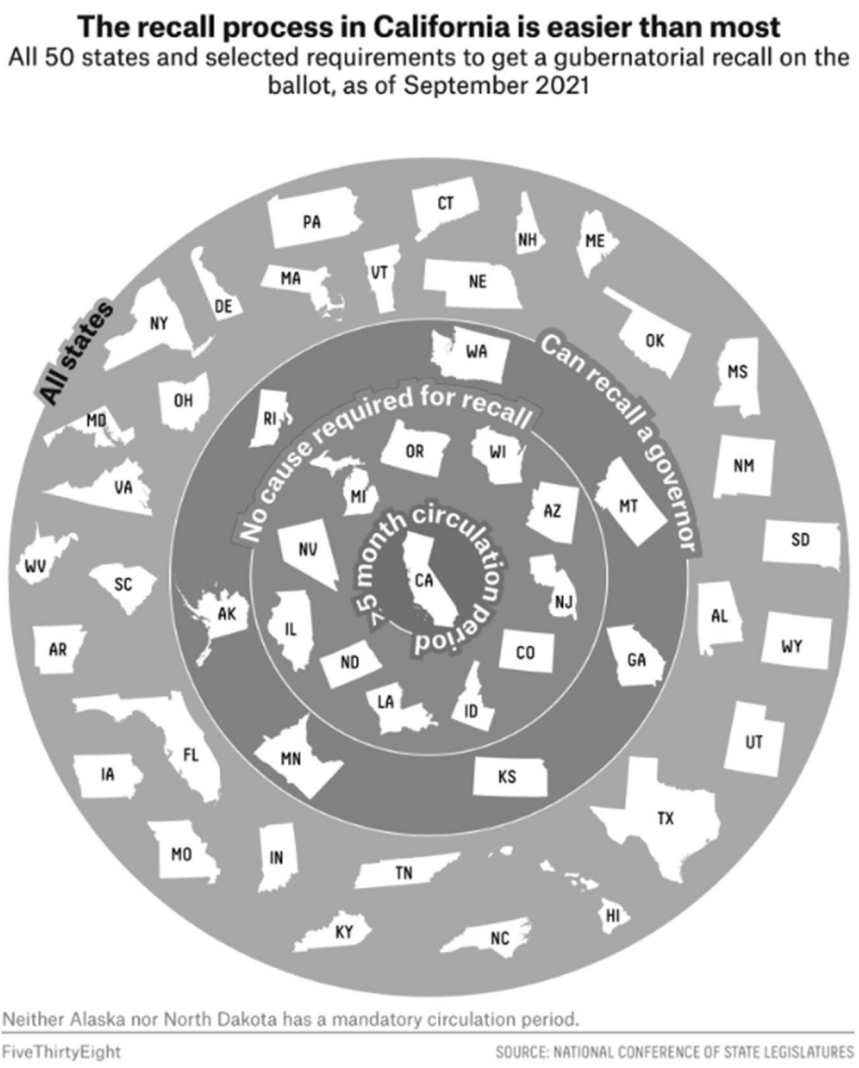


Fig. 18.12 An example of using color, shades, and different size of circles to categorize data. Source: Mithani & Sweedler (2021)

Linguistic Rhetoric

The effects of texts in data visualizations have been underrated. However, text formatting and language usage in data visualizations could also shape authors' intended meaning. Titles, subtitles, labels, and annotations on a chart could also influence audiences' interpretation of the visualization. Hullman and Diakopoulos (2011) proposed four linguistic rhetoric techniques that have been seen in data-informed news stories.

Typography
Texts are integral to data visualizations and function at multiple editorial levels (Moroianu, 2021). Besides ensuring the readability and legibility of texts, authors can also create visual hierarchy through typography. The choice of typeface, text color, size, shape, and boldness can all serve to convey the author's meaning. For example, by choosing the size of the text, authors decide what audiences should see first, second, and last (Muth, 2022). Text formatting can be used to signal importance, structure, or performance in data visualizations. For example, choosing different typefaces or sizes for titles, headings, subheadings, and body can help audiences quickly pick up the structure of the chart and eliminate confusion.

Irony
As a linguistic and literary device, irony is used where there is a contrast between literal meaning and implied meaning. Using irony in visualization titles can effectively attract audiences' attention. Using irony to phrase a rhetorical question implies its inverse and guides audiences to arrive at an obvious interpretation. For example, in Fig. 18.10, the subtitle includes a rhetorical question "Is there a lot of discrimination against Black people, or not?", intriguing audiences to look deeper into the below dot pie charts and draw their conclusions.

Similarity
Similarity techniques are applied in comparison situations where two entities are assumably similar. Authors can use analogy to explain the unclear meaning of two entities by comparing with two well-known entities (Hullman & Diakopoulos, 2011). Authors can also use the metaphoric statement to directly assert that one entity is the other, as in the data visualization (see Fig. 18.14) titled "Least Vaccinated U.S. Counties Have Something in Common: Trump Voters". Another similarity technique is Parallelism, displaying two similar statements or visual features to show they are equally important. For example, still in Fig. 18.13, the juxtaposition of two similar scatterplots is framed around the relationship between people's political stances and their vaccination hesitancy.

Individualization
The individualization techniques aim to directly appeal to the audience as an individual, which can increase the interest and the sense of participation of the audience (Hullman & Diakopoulos, 2011). One subtle means to achieve that is to provide the audience with exploratory functions in data visualization, such as sorting and filtering methods. For example, in Fig. 18.14, the data visualization provides a filter method that allows users to select the state they are interested in to explore the situation of administered doses in each state. Another popular means of individualization is to provide an interactive format of question and answer, where the story moves forward based on the audiences' answers to the questions.

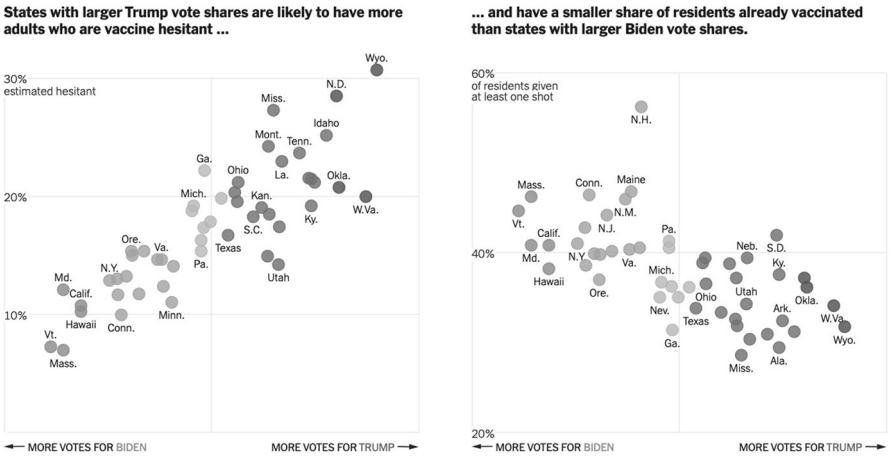


Fig. 18.13 Least Vaccinated U.S. Counties Have Something in Common: Trump Voters. An example of juxtaposing two similar scatterplots to represent the relationship between people's political stances and their vaccination hesitancy. Source: Ivory, Leatherby & Gebeloff (2021)

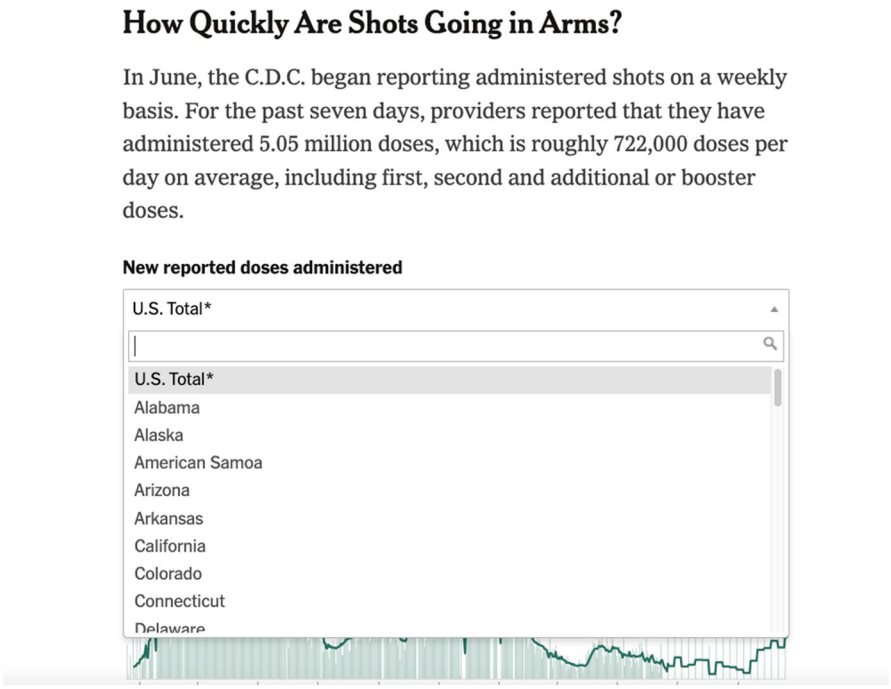


Fig. 18.14 An example of providing a filter method that allows users to select the state they are interested in to explore the situation of administered doses in each state. Source: The New York Times. See how vaccinations are going in your county and state

Conclusion

This chapter overviews previous literature on data storytelling, narrative visualization, and visualization techniques to offer a clear methodological view of how data and digital journalists can incorporate these strategies and techniques to improve data-driven storytelling. Nevertheless, the suggestions mentioned in this chapter do not intend to be either universal or prescriptive. The choice of narrative visualization structures, scrollytelling techniques, or linguistic techniques depends on many factors including the type of news story being prepared, the publishing platform, and the topic. For example, if a journalist is writing a breaking news story reporting the mortality rate in an earthquake, the Inverted Pyramid is better than any type of structure abovementioned. If the story is the result of a long-term, thorough investigation of a particular topic and integrates multiple datasets to conduct extensive analysis and investigation, then structures such as block stacks, Martini glass, and new water tower structures may be more appropriate. Therefore, not any data-driven story should go with the water tower or Martini structure, journalists need to consider multiple factors that might influence how a story unfolds when selecting the most appropriate structure.

Similarly, when applying scrollytelling techniques, journalists need to consider the time, cost, and labor before creating scrollers for a long-form story. Although the scrollytelling technique has gained popularity in newsrooms, it is labor-intensive and expensive.

How to use emerging technology to improve data-driven storytelling and engage audiences is a broad topic involving journalists, design, disseminating platforms, and audiences. This chapter concentrates only on the niche of visualization design and technique. Given that audiences have more agency in exploring the narrative of a news story, future studies can discuss more on how audiences engage in different structured stories and interact with different medium interfaces. As technology advances, data-driven storytelling continues to evolve, and journalists or designers continue developing new approaches to support stories with data visualization and better inform the public. In the context of future journalism, scholars need to continue discussing how the latest technology will transform the formats of journalism storytelling and professional practices, change the role of audiences in news consumption and production, and the relationship between journalists and audiences.

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